

Internship Based Project Report On

CUSTOMER REGION WISE DEMAND & SALES

ANALYTICS SYSTEM FOR RINL

A report submitted as a part of the industrial Orientation Training in IT & ERP
Department. Visakhapatnam Steel Plant.

By

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RASHTRIYA ISPAT NIGAM LIMITED (RINL)

Visakhapatnam Steel Plant

Duration – 4 Weeks (18th May 2026 to 13th June 2026)

CERTIFICATE

This is to certify that the following Students of **MAHARAJA VIJAYARAM GAJAPATHI RAJ COLLEGE OF ENGINEERING** are engaged in the project work titled A Project Based on **CUSTOMER REGION WISE DEMAND & SALES ANALYTICS SYSTEM FOR RINL** from 18th May 2026 to 13th June 2026.

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In partial Fulfilment of the degree **BACHELORS OF TECHNOLOGY** in **COMPUTER SCIENCE ENGINEERING & INFORMATION TECHNOLOGY** stream in **MAHARAJA VIJAYARAM GAJAPATHI RAJ COLLEGE OF ENGINEERING** is a record of Bonafide work carried out by an under the guidance and supervision during the period from 18th may 2026 to 13th June 2026.

Date:

Place: Visakhapatnam

S. NARASINGA RAO

Deputy General Manager (DGM),

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RINL-VSP

ACKNOWLEDGEMENT

First of all, I thank God in words for his grace, who gave us the opportunity and strength to carry out this work. The success and outcome of this project required a lot of guidance and assistance from many people and we are incredibly privileged to have got this all along with the completion of our project.

I thank to having this opportunity to thank Mr.S.NARASINGA RAO, IT and ERP Department. Visakhapatnam Steel Plant for guiding us with his immense knowledge and helping us to complete this project successfully.

I wish to express our sincere thanks to IT and ERP Dept. employees of Visakhapatnam Steel Plant for their valuable guidance in completing this project in time

Successfull

DECLARATION

I hereby that this project work entitled A Project Based on **CUSTOMER REGION WISE DEMAND & SALES ANALYTICS SYSTEM FOR RINL** is original and has not been submitted to any university or college before fulfilment of the requirements of any course of study or to the award of any degree. The opinion is given and the conclusions arrived at are of my own. The views expressed in the report do not represent the views of organization.

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ABSTRACT

The Customer Region-Wise Demand & Sales Analytics System for RINL is a web-based application designed to manage, analyze, and visualize sales data across different regions and states. Developed using Python, Flask, MySQL, HTML, CSS, and Chart.js, the system provides an efficient platform for storing and managing sales records while ensuring easy access to business information.

The application includes features for adding, viewing, searching, and updating sales records stored in a MySQL database. It enables users to monitor sales activities and customer demand patterns through an intuitive interface. By organizing data region-wise and state-wise, the system helps users gain a clear understanding of sales distribution and market trends.

The system shows sales data using charts, dashboards, and reports. It helps users understand customer demand and sales performance in different regions. This information supports better business planning and decision-making.

INTRODUCTION TO VSP

A BRIEF OVERVIEW:

Visakhapatnam Steel Plant (VSP) is the integrated steel plant Rashtriya Ispat Nigam Limited in Visakhapatnam which is founded in 1971, VSP strikes every one with a tremendous sense of awe, wonder and amazement as it presents a wide array of excellence in all its facets in scenic beauty, in management and above all in product quality. On the coast of Bay of Bengal and by the side of scenic Gangavaram beach, have risen tall and huge structures of technological architecture, the Visakhapatnam Steel Plant. But the vistas of excellence do not rest with the inherent beauty of location over sophistication of technology- they march ahead parading one aspect after another.

The decision of the Government of India to set up an integrated steel plant at Visakhapatnam was announced by then Prime Minister Smt. Indira Gandhi in Parliament on 17th January 1971. VSP is the first coastal based integrated steel plant of India is located, 16km west of city of destiny, Visakhapatnam, bestowed with modern technologies, VSP has an installed capacity of 3 million tonnes per annum of liquid steel and 2.656 million tonnes of saleable steel. The saleable steel here is in the form of Wire rod coils, Structural, Special Steel, Rebars, Forged

Rounds and etc. At VSP there emphasis on total automation, seamless integration and efficient up gradation, which result in wide range of long and structural products to meet stringent demands of discerning customers with India & abroad. VSP product meet exalting international Quality Standards such as JIS, DIN, BIS, BS etc. Rashtriya Ispat Nigam Limited, the corporate entity of Visakhapatnam Steel Plant is a Navaratna Public Sector Enterprise under the Ministry of Steel. Visakhapatnam Steel Plant is also fondly called as Vizag Steel. It is the first shore based Integrated Steel Plant in the country and is known for it's quality products delighting the customers. It is market leader in long products and it caters to the needs of diverse industrial sectors. It is the first steel plant to be certified ISO9001:2008 (presently 2015), ISO 14001:2004(presently 2015), OHSAS 18001:2007 and ISO/IEC 27001:2013 Standards. It is also the first PSE to be certified ISO 50001:2011- Energy Management Systems and has acquired CMMI Level 3 certification for S/W development. On 20th Jan 71 the then Prime minister of India laid the foundation stone of this plant. A new company Rashtriya Ispat Nigam Limited (RINL) was formed on 18th Feb 1982. From 1987 to 1992 various units of VSP were commissioned. VSP's Product Mix comprises Wire Rods, Bars, Angles, Channels/Beams, Rounds and Billets. The Plant also produces Pig Iron, Granulated Slag and Coal Chemicals. The rolled products find extensive usage in the Construction, Infrastructure, Railways, Power, Defence. Transport and Ship Building sectors. Coir's and Rods are used mainly for

reinforced concrete work for housing, construction of dams, buildings factories, manufacture of agricultural implements, fabrication of light engineering components. The Wire rods are used in Wire Drawing industry for electrodes transmission lines etc. The structural find application in engineering, house building, agricultural implements machinery, transmission towers, etc. VSP is having the state of the art technology.

The steel plant has many technological features, which are unique amongst the steel plants in the country. The company is a pioneer in introducing many new technologies in the country. The production of TMT rebars by temporal process is a shining example in this respect.

VSP has become the first integrated steel plant in the country to be certified to all the three international standards for quality (ISO-9001), for Environment Management (ISO) -14001) & for Occupational Health & Safety (OHSAS 18001). The certificate covers quality systems of all operational, maintenance, service units besides purchase systems, Training and Marketing functions spreading over 4 Regional Marketing offices, 20 branch offices & 22 stock yards located all over the country. VSP by successfully installing & operating efficiently Rs. 460 process worth of pollution and Environment Control Equipment and converting the barren landscape more than 3 million plants has made the steel plant, steel Township and VSP exports Quality Pig Iron & Steel products Sri Lanka, Myanmar, Nepal, Middle East, USA & South East Asia (Pig Iron). RINL-VSP was awarded "Star

Trading House" status during 1997-2000. Having established a fairly dependable export market, VSP plans to make a continuous presence in the export market.



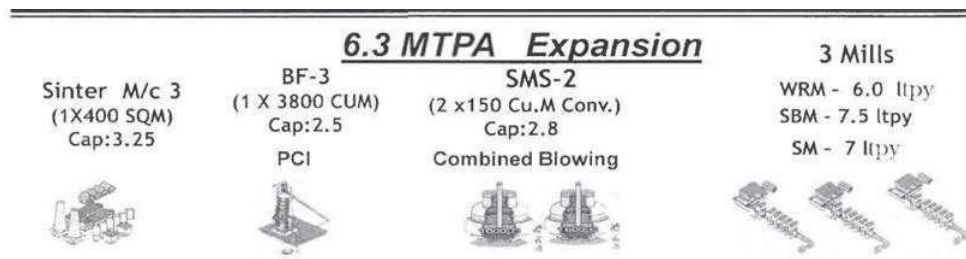
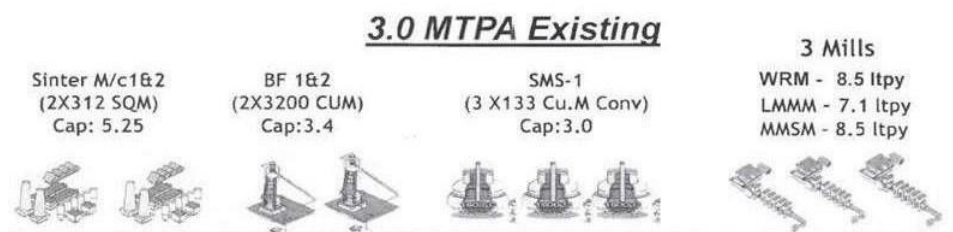
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Different sections at the RINL VSP:

- Coke oven and coal chemicals plant
- Sinter plant
- Blast Furnace
- Steel Melt Shop
- Continuous casting machine
- Light and medium machine mills
- Calcining and refractive materials plant
- Rolling mills

- Thermal power plant Chemical power plant
- Yard management

The expansion of production facilities from 3MTPA stage to 7.3 MTPA stage has been shown below:



SINTER PLANT

Sinter plants agglomerate iron ore fines (dust) with other fine materials at high temperature, to create a product that can be used in a blast furnace. The final product, a sinter, is a small, irregular nodule of iron mixed with small amounts of other minerals. The process, called sintering, causes the constituent materials to fuse to make a single porous mass with little change in the chemical properties of the ingredients. The purpose of sinter are to be used converting iron into steel.

Sinter plants, in combination with blast furnaces, are also used in non-ferrous smelting. About 70% of the world's primary lead production is still produced using the sinter plant–blast furnace combination,[1] and this combination was formerly often used in copper smelting (at the Electrolytic Refining and Smelting smelter in Wollongong, New South Wales,

INFORMATION TECHNOLOGY & ERP DEPARTMENT

VSP is the first Steel Plant in India where integrated Automation and computerization is envisaged right from the beginning. Almost all the production and business processes are controlled by computer in one form or the other. These computers and the related automation systems are meant to aid all levels of personnel involved in operation and maintenance of the plant. They provide valuable information for the top management to take both long and short-term decisions. IT applications at RINL have been developed and implemented keeping the overall organizational business objectives in view. IT infrastructure has been upgraded recently with IBM-RS 6000 servers and ORACLE-91 as the database server and data communication with backbone Fiber- optic network. In the area of marketing VPN based wide area networks has been implemented to provide vital information to regional marketing offices as well as customers. **VSP** has a sizable amount of computerization built into its design in line with the latest trends in technology. The main objective of computerization in **VSP** is to assist the operators in controlling repetitive jobs, to optimize utilization of scarce resources by real time process control and assist in operation and production management. The computerization and automation in **VSP** is envisaged in three levels for effective integration between planning and processes. The highest level assists management decision-

making by collection, analysis and providing the production and management information on a real time basis and the lowest level assists the actual process operators by reducing their tedium in the form of automation. Thus the structure is designed for assisting all levels of operation and management of the plant.

BUSINESS APPLICATION SYSTEMS:

This is for top level of the hierarchy and is consisting of a Central Computer. All the large online business applications covering different business functions of the plant are implemented in this computer. This generates all the information/reports for the management control. It also stores and provides long-range plant operating, maintenance, commercial and finance data. Based on the long-range information this computer plans the production scheduling for the shops. It generates data regarding the products to be produced, maintenance plans and dispatch plans. These production and maintenance plans are transmitted to the concerned process control computers/personnel. Subsequently the actual plant operational, maintenance data are collected. Some of the important business applications running in VSP are listed below:

1. Payroll System
2. Costing System

3. Financial Accounting System
4. Marketing System
5. Attendance Recording System
6. Library Information System
7. Materials Management System
8. Maintenance Management System
9. Production Planning and Control System
10. Wagon Information Tracking System
11. On-line TC printing System
12. Human Resource Information System
13. Raw Material Management System
14. Coke Ovens Heating and Regulating

SYSTEM PROCESS CONTROL SYSTEMS :

The main jobs of the process control computers are memorizing production schedules given by zonal computers, monitoring the shop production and the material flow in the shops and storing and execution of process control parameters and algorithms to control the process. They also control the lower level automation systems. The most important aspect of these computers is the man machine dialogue envisages. Such

computer systems are envisaged in the Blast Furnaces, Steel Melt Shop, Continuous Casting Machines & Rolling Mills. Apart from these computers, there is a 'Supervisory Control And Data Acquisition (SCADA)' computer to monitor and control Gas, Power, Water and their distribution in the plant.

FIELD LEVEL AUTOMATION SYSTEMS:

This level consists of local automation systems. These systems do logic, sequence, loop control and indication of process parameters to operation personnel in the plant for taking up timely corrective actions to achieve the quality production. This level consists of Programmable Logic Control (PLC) Systems, analog (split architecture type) and microprocessor-based instrumentation systems. The above concept of hierarchical and integrated computer systems in line with the latest technology is introduced in VSP for the first time in India. This system is contributing a lot in achieving the quality and productivity of steel with an integrated control of production and processes. Thus, VSP is a trendsetter in computerization in basic industries in India with latest systems and opening new vistas in computerization in our country.

INTRANET WEB SITE :

An intranet web site is also maintained by VSP IT Department and many facilities are hosted in this site for the convenience of the different sections of the plant and for the benefit of the employees. The content in this site is

updated regularly. There are many important features/modules hosted in this site.

NETWORK

VSP is having a Gigabit Ethernet Data Communication Network, working on Optical Fiber Cable(OFC) network. This supports data transfer at the rate of 1000 Mbps on all the main links. The campus network is divided into eleven zones. These zones are inter-linked to the central core switch using Single mode type optical fibers. The Central core switch is located at Central Computer Centre. The 11 Zonal switches are located at Coke Ovens, Sinter Plant, SMS, Central stores, BC Gate, LMMM, MMSM, Central computer centre, Admin building, Township and General Hospital. Further, a total of 200 access switches are connected to the network through the 11 zonal switches, using Multimode/ single mode types fiber optical cable,. The access switches provide connectivity to approximately 3000 nodes throughout the plant premises, Administration building area and township. ERP It stands for Enterprise Resource Planning. ERP refers to a suite of software that organizations use to manage day-to-day business activities, such as accounting, procurement, project management, risk management, compliance, and supply chain operations. In Visakhapatnam Steel Plant ERP is being used extensively and it is covering many applications and most of the departments.

INTRODUCTION TO THE PROJECT

In many organizations, sales data is stored in large datasets, making it difficult to quickly identify customer demand patterns and sales performance across different regions. Manual analysis is time-consuming and may not provide clear insights for decision-making.

To overcome this problem, the **Customer Region-Wise Demand & Sales Analytics System for RINL** was developed. The project helps collect, manage, and analyze sales data efficiently through a centralized dashboard. It provides visual charts, reports, and analytics that help identify high-demand regions, top-performing products, and sales trends, enabling better business planning and decision-making.

This project is built using **Python, Flask, MySQL, HTML, CSS, and Chart.js**. It provides an interactive dashboard with visual reports, sales data management, search functionality, and sales analytics charts. The system enables users to add, view, update, and analyze sales records efficiently, improving the overall monitoring and management of sales activities.

The main objective of the project is to transform raw sales data into meaningful information through dashboards, charts, and reports, thereby supporting better decision-making and business growth.

- 1) Frontend (HTML, CSS, Chart.js)
- 2) Backend (Python, Flask, MYSQL)

HTML:

The Hypertext Markup Language, or HTML is the Standard Markup Language for documents designed to be displaying in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages as JavaScript. Web Browsers receive HTML documents from a webserver or from local storage and render the documents into multimedia webpages. HTML describes the structure of a webpage semantically and originally includes cues for appearance of document.

CSS:

Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation of a document written in a markup language such as HTML. CSS is a corner stone technology of the World Wide Web, alongside HTML and JavaScript. CSS is designed to enable the separation of presentation and content, including layout, colors and fonts. This separation can improve content accessibility; provide more flexibility and control in the specification of presentation characteristics; enable multiple webpages to share formatting by specifying the relevant CSS in a separate .css file, which reduces complexity and repetition in the structural content; and enable the .css file to be cached to improve the page load speed between the pages that share the file and its formatting.

PYTHON FLASK:

Flask is a lightweight web framework for Python used to build web applications, APIs, and dynamic websites. It is simple, flexible, and easy to learn, making it popular among beginners and professionals.

Flask is based on the WSGI toolkit and Jinja2 template engine. It provides the basic tools needed for web development while allowing developers to add extensions as required.

CHARTS.JS:

Chart.js is a popular open-source JavaScript library used to create interactive and responsive charts for web applications. It allows developers to visualize data using various chart types with minimal code.

Features

- Simple and easy to use
- Responsive charts
- Interactive animations
- Customizable colors and styles
- Supports multiple chart types

Supported Chart Types

- Bar Chart
- Line Chart
- Pie Chart
- Doughnut Chart
- Radar Chart
- Polar Area Chart
- Scatter Chart
- Bubble Chart

MYSQL:

MySQL is currently the most popular database management system software used for managing the relational database. It is open-source database software, which is supported by Oracle Company. It is fast, scalable, and easy to

use database management system in comparison with Microsoft SQL Server and Oracle Database. It is commonly used in conjunction with PHP scripts for creating powerful and dynamic server- side or web-based enterprise applications.

It is developed, marketed, and supported by MySQL AB, a Swedish company, and written in C programming language and C++ programming language. The official pronunciation of MySQL is not the My Sequel; it is My Ess Que Ell.

However, you can pronounce it in your way. Many small and big companies use MySQL. MySQL supports many Operating Systems like Windows, Linux, MacOS, etc. with C, C++, and Java languages.

MySQL is a Relational Database Management System (RDBMS) software that provides many things, which are as follows:

- It allows us to implement database operations on tables, rows, columns, and indexes.
- It defines the database relationship in the form of tables (collection of rows and columns), also known as relations.
- It provides the Referential Integrity between rows or columns of various tables.
- It allows us to updates the table indexes automatically.
- It uses many SQL queries and combines useful information from multiple tables for the end-users

DESIGN OF PROJECT AND DATABASE SCHEMA

The below is the project overview and the design which was made before writing and developing the code for the website.

Project Modules Overview

This portal is divided into five core modules:

- **Dashboard Module**
- **Sales Entry Module**
- **View Sales Module**
- **Edit Module**
- **Heatmap Module**
- **Analytics and Visualization Module**
- **Reports Module**
- **Database Module**

UI/UX Design

The **UI/UX (User Interface/User Experience) Design** of the Customer Region-Wise Demand & Sales Analytics System is developed to provide a simple, user-friendly, and interactive experience for users. The interface is designed using **HTML, CSS, and Chart.js** to ensure easy navigation and effective data visualization.

breakdown of each:

UX Design (User Experience Design)

Definition:

User Experience (UX) design is the process of enhancing user satisfaction by improving the usability, accessibility, and pleasure provided in the interaction between a user and a product or service. It's about understanding what users need, what their goals are, and designing a product that effectively helps them achieve those goals while making the experience as intuitive and enjoyable as possible.

Description:

Think of UX design as the **architectural blueprint** and the **invisible backbone** of a product. A UX designer is concerned with the *entire journey* a user takes when interacting with something. This includes:

- **Research:** Deeply understanding the target users through interviews, surveys, and analyzing their behaviors, needs, and pain points. They ask questions like: Who are the users? What are their motivations? What problems are they trying to solve?
- **Information Architecture (IA):** Organizing and structuring content in a logical and easy-to-find way. This is like creating the sitemap for a website or the navigation flow for an app, ensuring users can find what they're looking for intuitively.

- **Interaction Design (IxD):** Defining how users will interact with the product – what happens when they click a button, swipe, or enter text. It's about designing the "flow" and responsiveness of the system.
- **Wireframing and Prototyping:** Creating low-fidelity (basic outlines) and high-fidelity (more detailed, interactive models) representations of the product to test flows and gather early feedback before visual design is applied.
- **Usability Testing:** Observing real users interacting with prototypes or live products to identify areas of confusion, frustration, or inefficiency, and then iterating on the design based on that feedback.
- **Accessibility:** Ensuring the product can be used by people with diverse abilities and disabilities, adhering to guidelines that make it inclusive for everyone.

In essence, UX design is about the "why," "what," and "how" of product use. It focuses on solving user problems and ensuring the product is useful, usable, desirable, and valuable.

UI Design (User Interface Design)

Definition:

User Interface (UI) design is the process of designing the visual and interactive elements of a digital product's interface. It focuses on the "look and feel" – how the product is presented to the user, including the screens, buttons, icons,

typography, colors, and other visual components that users directly interact with.

Description:

If UX is the blueprint, then UI is the **interior design** and **decoration** of the product. A UI designer takes the structural and functional framework laid out by the UX designer and brings it to life visually. Their responsibilities include:

- **Visual Design:** Choosing and applying color palettes, typography, imagery, icons, and graphical elements to create a consistent and aesthetically pleasing visual identity for the product.
- **Layout and Grids:** Arranging all the visual elements on the screen in an organized and harmonious way, considering spacing, alignment, and visual hierarchy to guide the user's eye.
- **Interactive Elements:** Designing specific UI components like buttons, sliders, input fields, toggles, menus, and notifications, ensuring they are not only visually appealing but also clear, intuitive, and consistent in their behavior.
- **Branding and Consistency:** Ensuring that the visual design reflects the brand's identity and that design elements are consistent across all screens and platforms for a unified user experience.
- **Responsive Design:** Designing interfaces that adapt and look good on various screen sizes and devices (desktops, tablets, mobile phones).
- **Micro-interactions and Animations:** Adding subtle animations,

transitions, and feedback elements that make the interface feel alive, responsive, and delightful to use.

In essence, UI design is about the "aesthetics" and "interactivity" of the product. It focuses on making the product visually appealing, easy to navigate, and pleasurable to interact with.

The Relationship Between UI and UX:

While distinct, UI and UX are interdependent. A fantastic user experience can be undermined by a poor user interface, and a beautiful interface won't save a product that's difficult to use or doesn't meet user needs.

- UX comes first: Typically, UX research and strategy inform UI design. You figure out *what* needs to be built and *how* it should function (UX) before designing *how it looks* (UI).
- UI brings UX to life: The UI is the tangible manifestation of the UX. A well-designed UI makes the well-thought-out UX accessible and enjoyable.
- They collaborate: UI and UX designers often work closely together in a team, with the UX designer providing the foundational research and wireframes, and the UI designer crafting the visual and interactive details based on that foundation.

Together, UI/UX Design creates products that are not only functional and useful but also delightful and easy to use, leading to increased user satisfaction

Consistent Layout: Sidebar navigation, top bar with branding

Responsive Design: Mobile-friendly UI, table scrolling support

Accessible Colors: High contrast for readability

Loading Indicators: For data fetch operations

Dashboard Module:

- The Dashboard Module acts as the central interface of the system, presenting sales data in a visual and organized manner. It retrieves information from the database and displays key performance indicators through summary cards and charts. This helps users quickly understand sales trends, identify high-demand regions, and analyze overall business performance without manually examining large datasets.
- By presenting data in a visual and organized manner, the Dashboard Module reduces the effort required for manual analysis and supports faster business decision-making. It acts as the primary interface for monitoring sales activities and evaluating organizational performance.

Purpose:

The Dashboard Module provides a quick overview of sales performance and customer demand. It helps users monitor important business metrics and make informed decisions based on real-time data.

Features:

Dashboard:fig

Required fields:

- Displays Total Sales
- Shows Total Number of Orders
- Identifies Highest Demand Region
- Displays Top Performing Product
- State-wise and Region-wise Sales Analysis
- Interactive Charts and Visualizations
- Dynamic data retrieval from MySQL database
- User-friendly dashboard interface

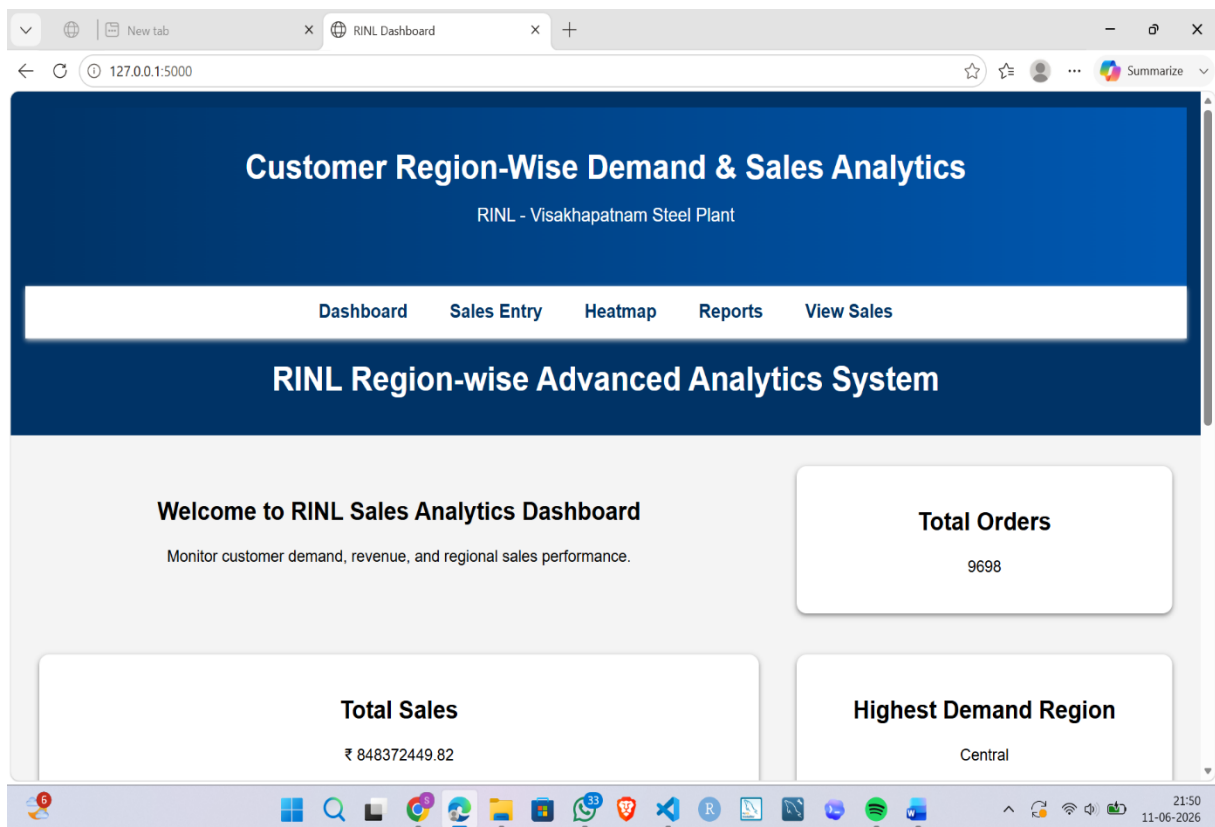
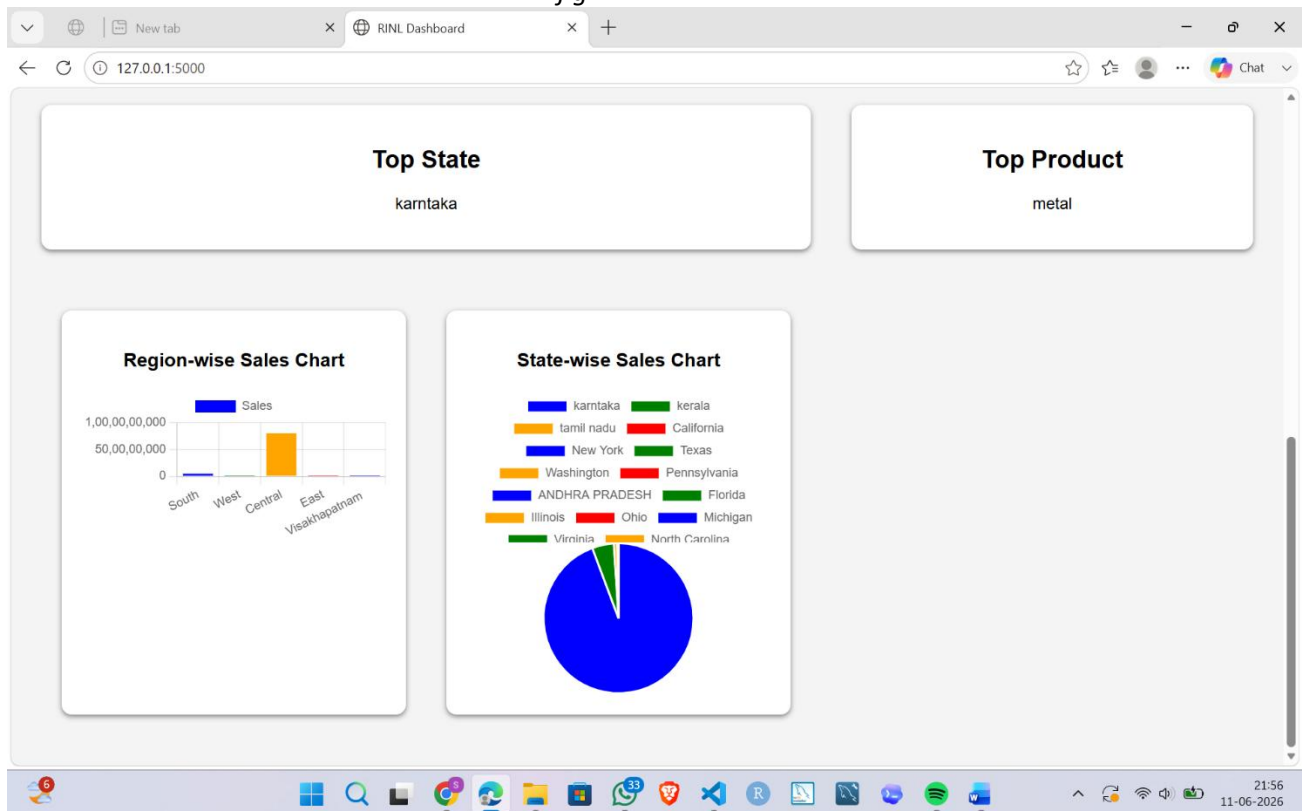


Fig:1

The Dashboard serves as the main page of the Customer Region-Wise Demand & Sales Analytics System. It retrieves data from the MySQL database and presents it in an easy-to-understand format using summary cards and charts.

The dashboard displays key performance indicators such as Total Sales, Total Orders, Top Product, Highest Demand Region, and State-wise Sales Analysis. It enables users to quickly understand business performance without manually analyzing large datasets.

fig:2



SALES ENTRY MODULE:

The Sales Entry Module serves as the data collection component of the Customer Region-Wise Demand & Sales Analytics System. It allows users to enter details such as Order ID, Customer Name, State, Region, Product Name, and Sales Amount through a user-friendly interface. The entered information is processed by the Flask backend and stored securely in the MySQL database.

This module ensures that sales data is maintained accurately and consistently. Every new record added through the form becomes immediately available for analysis in the Dashboard, Reports, and Visualization modules. By providing a structured method for data entry, the module reduces manual errors and improves data reliability.

The Sales Entry Module plays a crucial role in the overall system as it acts as the primary source of information for customer demand analysis and sales performance monitoring. The quality and accuracy of the data entered directly influence the effectiveness of the analytics and reporting features available in the application.

PURPOSE:

The Sales Entry Module is designed to capture and store sales-related information in the database. It ensures that sales data is collected systematically and made available for further analysis, reporting, and decision-making.

Browser tabs: New tab, Sales Entry

Address bar: 127.0.0.1:5000/sales

Form fields:

- Order ID
- dd-mm-yyyy
- Customer Name
- State
- Region
- Product Name
- Sales Amount
- Quantity
- Profit

Add Sales Record

Taskbar: 22:10 11-06-2026

fig3

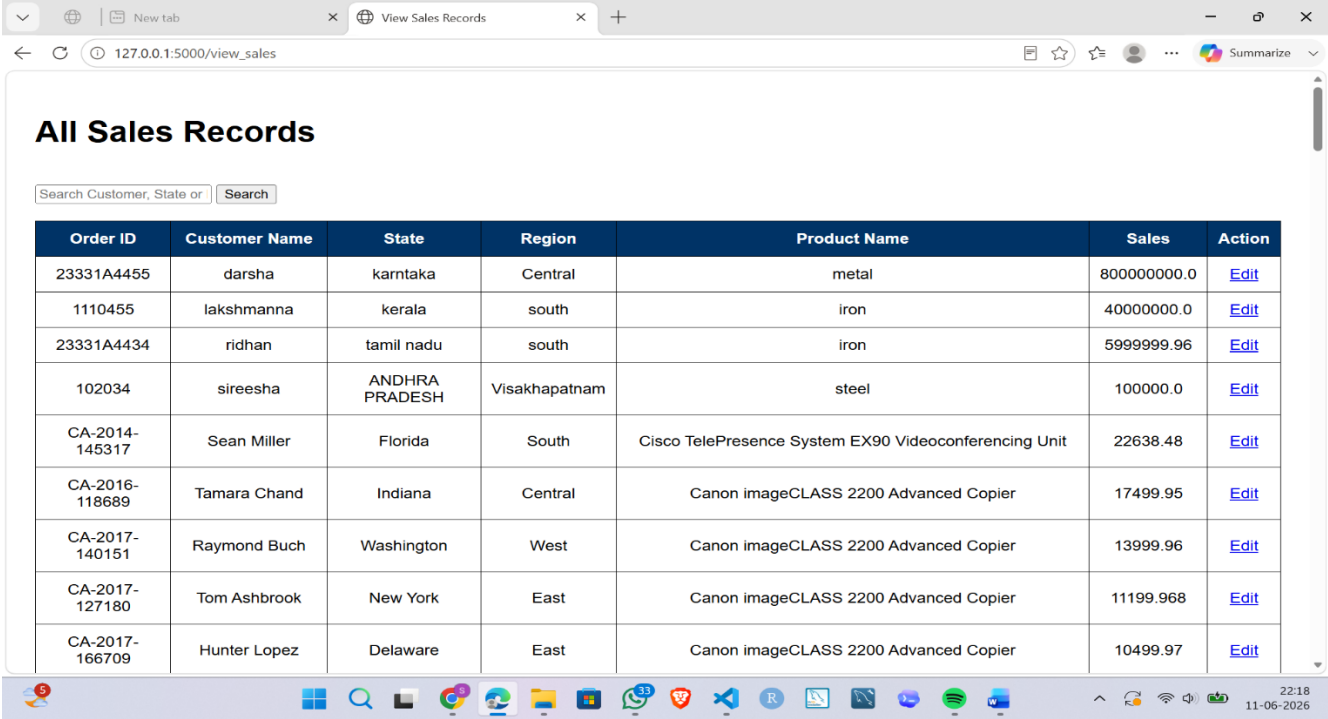
VIEW SALES MODULE:

Purpose:

The View Sales Module is used to display all sales records stored in the database. It allows users to access, review, and manage sales information in a structured and organized manner.

Features:

- Displays all sales records in a tabular format
- Retrieves data dynamically from the MySQL database
- Provides a clear and organized view of sales information
- Supports record searching and filtering
- Enables editing of existing records
- Facilitates efficient sales data management



All Sales Records

Search Customer, State or

Order ID	Customer Name	State	Region	Product Name	Sales	Action
23331A4455	darsha	karntaka	Central	metal	800000000.0	Edit
1110455	lakshmanna	kerala	south	iron	40000000.0	Edit
23331A4434	ridhan	tamil nadu	south	iron	5999999.96	Edit
102034	sireesha	ANDHRA PRADESH	Visakhapatnam	steel	100000.0	Edit
CA-2014-145317	Sean Miller	Florida	South	Cisco TelePresence System EX90 Videoconferencing Unit	22638.48	Edit
CA-2016-118689	Tamara Chand	Indiana	Central	Canon imageCLASS 2200 Advanced Copier	17499.95	Edit
CA-2017-140151	Raymond Buch	Washington	West	Canon imageCLASS 2200 Advanced Copier	13999.96	Edit
CA-2017-127180	Tom Ashbrook	New York	East	Canon imageCLASS 2200 Advanced Copier	11199.968	Edit
CA-2017-166709	Hunter Lopez	Delaware	East	Canon imageCLASS 2200 Advanced Copier	10499.97	Edit

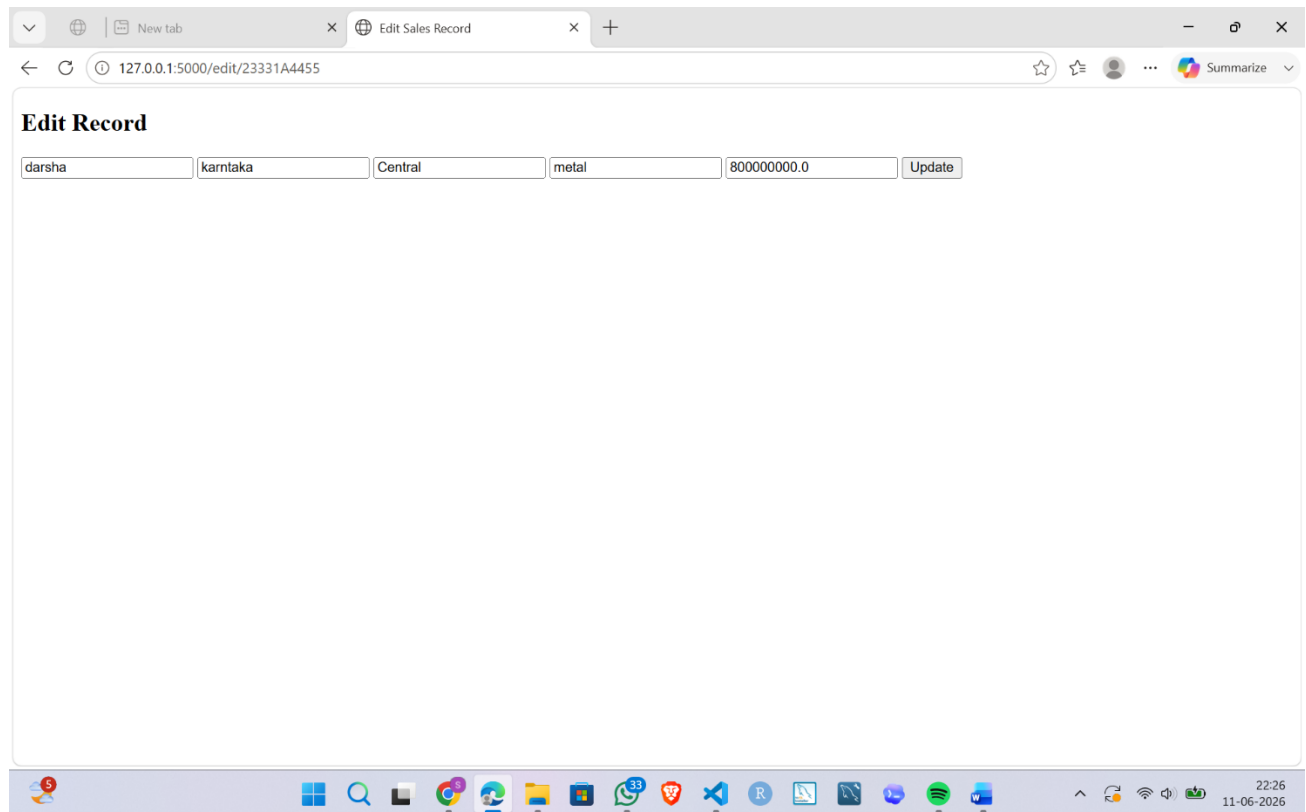
EDIT MODULE:

The Edit Module allows users to modify existing sales records stored in the database. It helps maintain accurate and up-to-date sales information by enabling corrections and updates whenever required.

Features:

- *Edit existing sales records*
- *Retrieve record details based on Order ID*
- *Update customer, product, region, state, and sales information*
- *Save changes directly to the MySQL database*
- *Redirect users to the View Sales page after successful update*
- *Improves data accuracy and consistency*

The Edit Module enables users to update sales information that has already been stored in the system. When a user selects a record for editing, the application retrieves the corresponding data from the MySQL database and displays it in a pre-filled form. The user can modify the required fields and submit the changes.



The updated information is processed through the Flask backend and stored in the database using SQL update operations. This module ensures that incorrect or outdated records can be corrected easily, thereby maintaining the reliability and accuracy of the sales data. It plays an important role in effective sales management and supports better decision-making through accurate analytics and reporting.

REPORTS MODULE:

Purpose:

The Reports Module is designed to generate summarized sales reports that help users analyze sales performance and customer demand across different regions and states. It provides meaningful insights for business monitoring and decision-making.

Features:

- Generates sales summary reports
- Displays region-wise and state-wise sales analysis
- Retrieves data dynamically from the MySQL database
- Helps identify high-performing regions and products
- Supports business decision-making through analytical insights
- Provides organized and easy-to-understand report formats

The Reports Module plays an important role in transforming raw sales data into useful business information. It collects data from the MySQL database and presents it in a structured format, making it easier for users to understand sales trends and performance patterns.

Region	Total Sales
South	46388983.54
West	713471.34
Central	800497800.87
East	672194.05
Visakhapatnam	100000.0

This module helps users analyze sales across different regions, compare performance, and identify areas with high customer demand. By providing detailed summaries and analytical reports, it supports strategic planning and improves decision-making. The Reports Module serves as a valuable tool for monitoring business growth and evaluating overall sales performance.

HEATMAP MODULE:

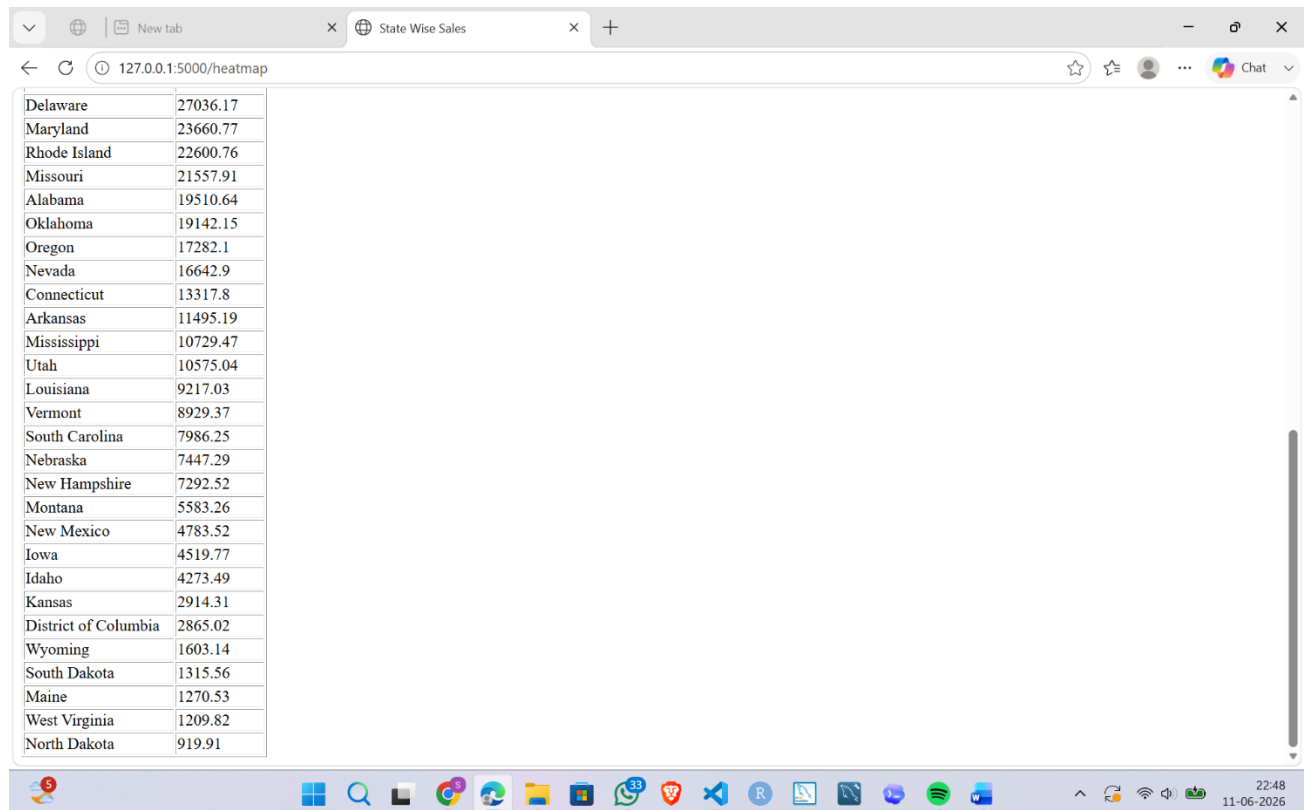
PURPOSE:

The Heatmap Module is used to visually represent sales performance and customer demand across different states and regions. It helps users quickly identify high-demand and low-demand areas through color-based analysis.

Features:

- Displays sales distribution using a heatmap visualization
- Identifies high-performing and low-performing regions
- Provides state-wise sales analysis
- Uses color intensity to represent sales values
- Retrieves data dynamically from the MySQL database
- Supports quick and effective decision-making

The Heatmap Module provides a visual representation of sales data by using different color intensities to indicate sales performance across various states and regions. Areas with higher sales are highlighted with stronger colors, while areas with lower sales are displayed with lighter colors. This enables users to easily identify customer demand patterns and regional sales trends.



The module retrieves sales information from the MySQL database and processes it through the Flask backend before displaying it in the heatmap format. By converting numerical data into a visual representation, the Heatmap Module simplifies data analysis and helps management focus on high-demand markets, improve resource allocation, and make informed business decisions.

Analytics and Visualization Module

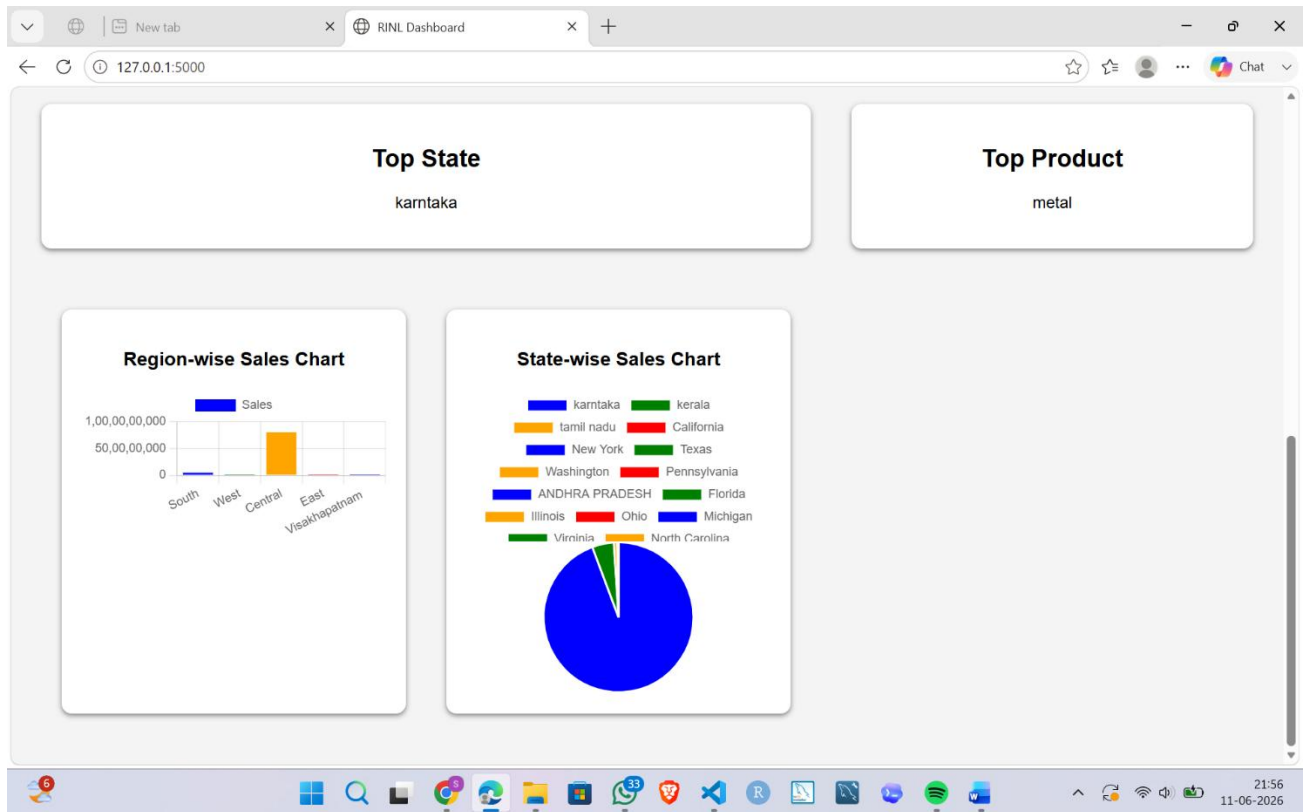
Purpose:

The Analytics and Visualization Module is designed to analyze sales data and present it in a graphical format. It helps users understand customer demand patterns, sales trends, and regional performance through interactive charts and visual reports.

Features:

- Region-wise Sales Analysis
- State-wise Sales Analysis
- Interactive Bar Charts and Pie Charts
- Dynamic data retrieval from MySQL database
- Visual representation of customer demand trends
- Easy identification of high-performing regions and products
- Supports data-driven decision-making

The Analytics and Visualization Module transforms raw sales data into meaningful insights using graphical representations. The module retrieves data from the MySQL database and displays it through charts and visual reports, making complex sales information easier to understand and analyze.



It helps users identify sales trends, compare performance across different regions and states, and determine areas with high customer demand. By presenting data visually, the module enhances business intelligence and supports effective decision-making. The use of charts and graphs allows users to quickly interpret sales performance and gain valuable insights without manually analyzing large datasets.

DATABASE MODULE:

Purpose:

The Database Module is responsible for storing, managing, and retrieving all sales-related data

used in the system. It acts as the central repository that supports data analysis, reporting, and

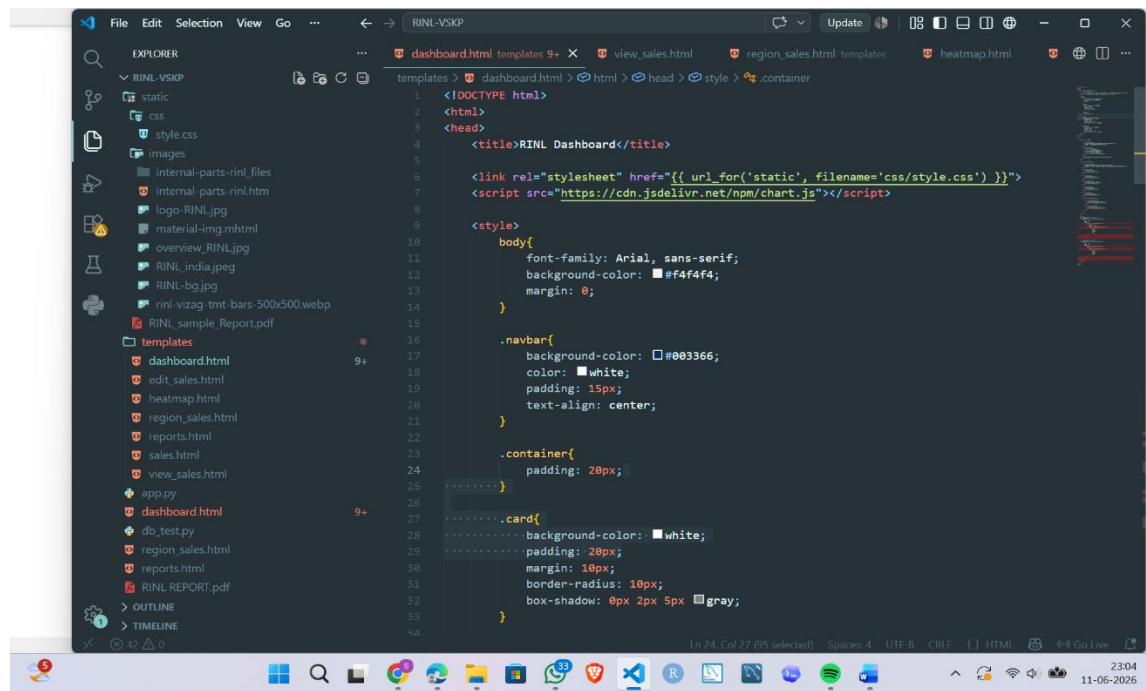
visualization.

Features:

- Stores sales records securely in MySQL
- Supports data insertion, retrieval, and updating operations
- Maintains customer, product, region, and sales information
- Ensures data consistency and accuracy
- Provides data for dashboards, charts, reports, and analytics
- Enables efficient data management and querying

Sample code

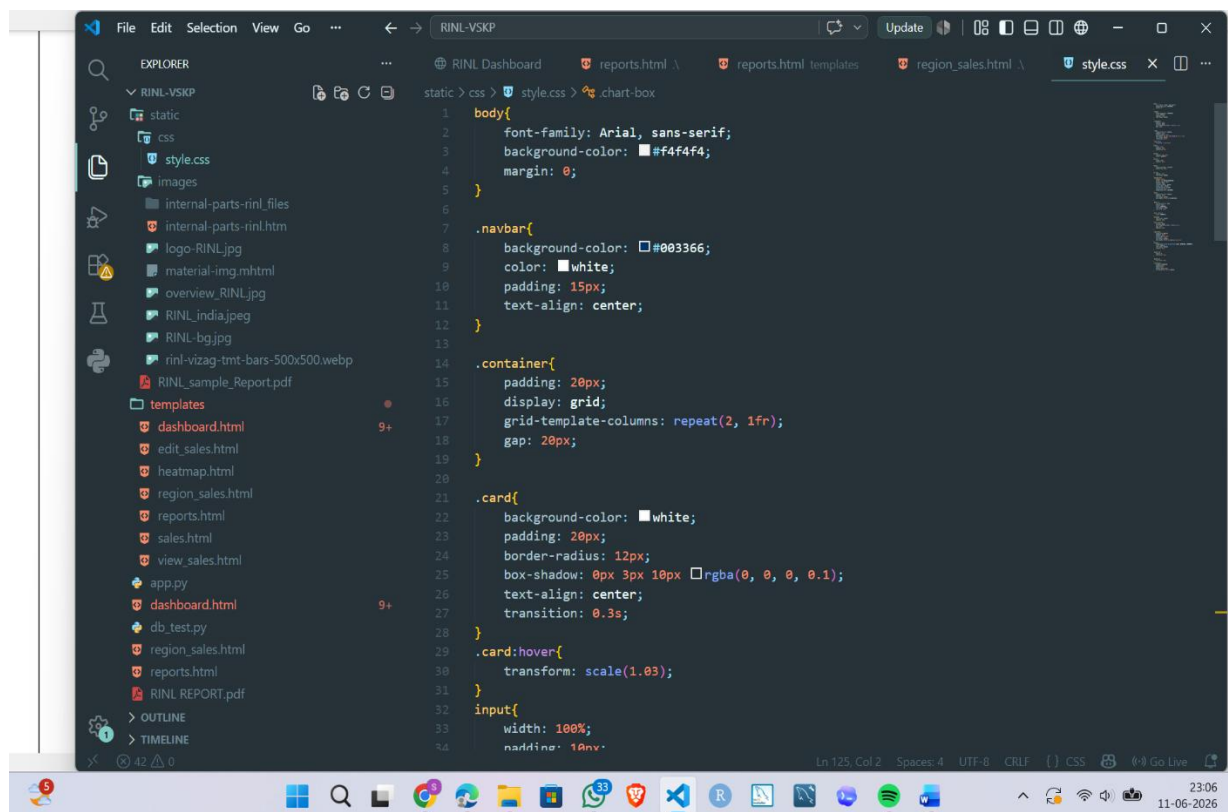
HTML:



The screenshot shows a Visual Studio Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'RINL-VSKP' with a 'templates' folder containing several HTML files. The code editor displays the 'dashboard.html' file, which is a template for a dashboard. The code includes a DOCTYPE declaration, a title 'RINL Dashboard', a link to a stylesheet, and a script for a chart. The CSS is embedded in the HTML file.

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>RINL Dashboard</title>
5
6   <link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
7   <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
8
9   <style>
10     body{
11       font-family: Arial, sans-serif;
12       background-color: #f4f4f4;
13       margin: 0;
14     }
15
16     .navbar{
17       background-color: #003366;
18       color: white;
19       padding: 15px;
20       text-align: center;
21     }
22
23     .container{
24       padding: 20px;
25     }
26
27     .card{
28       background-color: white;
29       padding: 20px;
30       margin: 10px;
31       border-radius: 10px;
32       box-shadow: 0px 2px 5px gray;
33     }
34
```

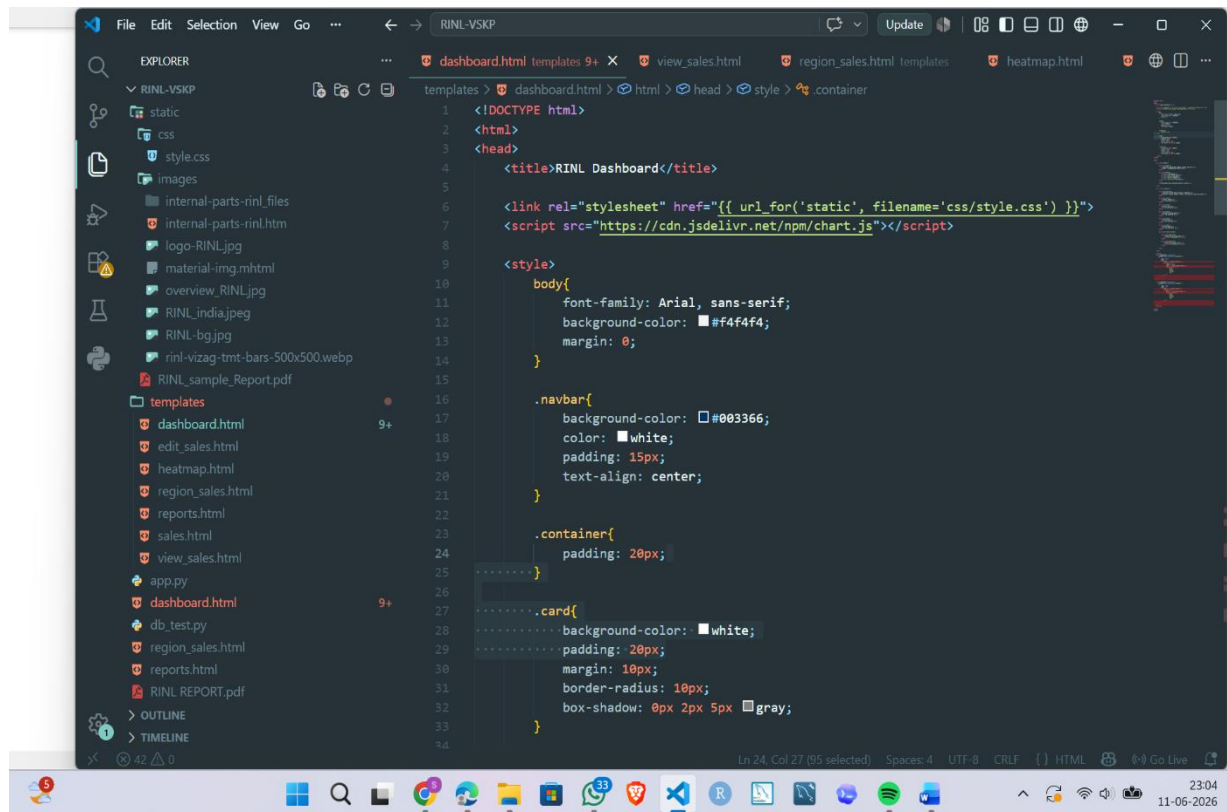
CSS:



The screenshot shows a Visual Studio Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'RINL-VSKP' with a 'static' folder containing a 'style.css' file. The code editor displays the 'style.css' file, which contains CSS rules for the dashboard. The rules include a body rule, a navbar rule, a container rule, a card rule, and an input rule.

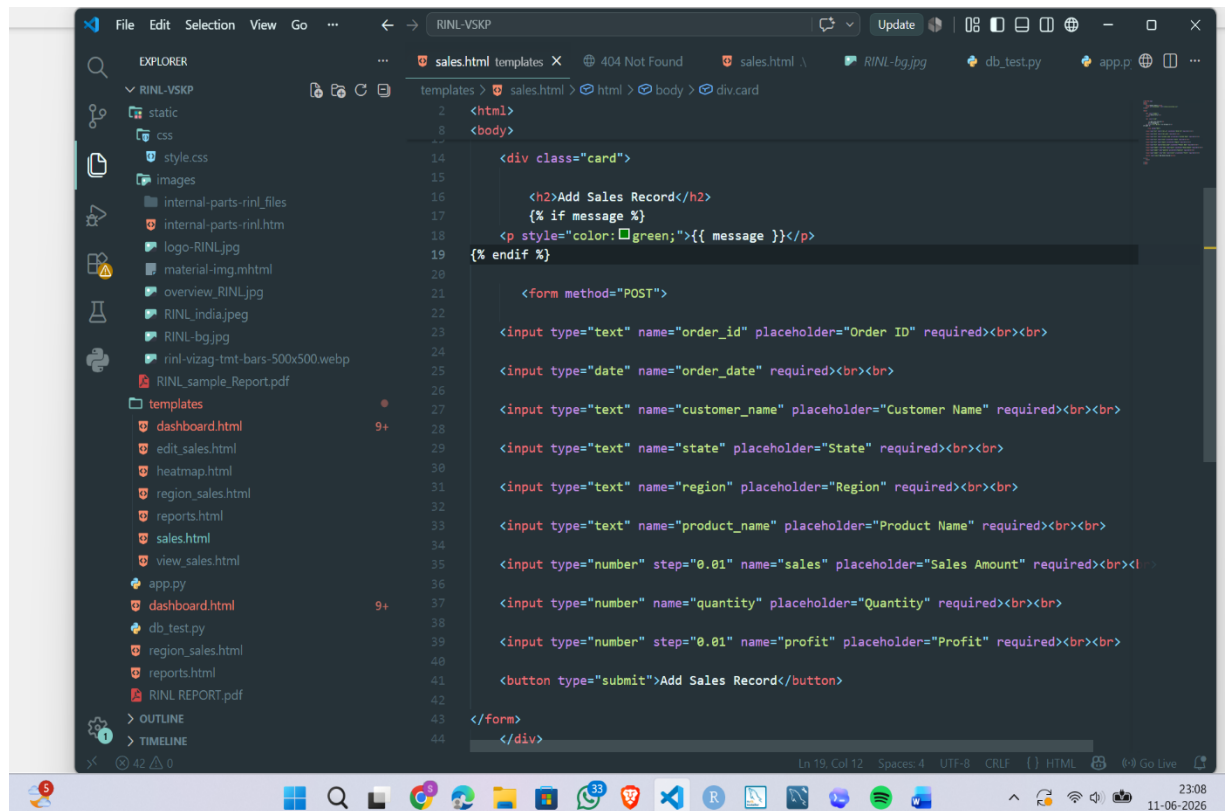
```
1 body{
2   font-family: Arial, sans-serif;
3   background-color: #f4f4f4;
4   margin: 0;
5 }
6
7 .navbar{
8   background-color: #003366;
9   color: white;
10  padding: 15px;
11  text-align: center;
12 }
13
14 .container{
15   padding: 20px;
16   display: grid;
17   grid-template-columns: repeat(2, 1fr);
18   gap: 20px;
19 }
20
21 .card{
22   background-color: white;
23   padding: 20px;
24   border-radius: 12px;
25   box-shadow: 0px 3px 10px rgba(0, 0, 0, 0.1);
26   text-align: center;
27   transition: 0.3s;
28 }
29
30 .card:hover{
31   transform: scale(1.03);
32 }
33
34 input{
35   width: 100%;
36   padding: 10px;
37 }
```

DASHBOARD:



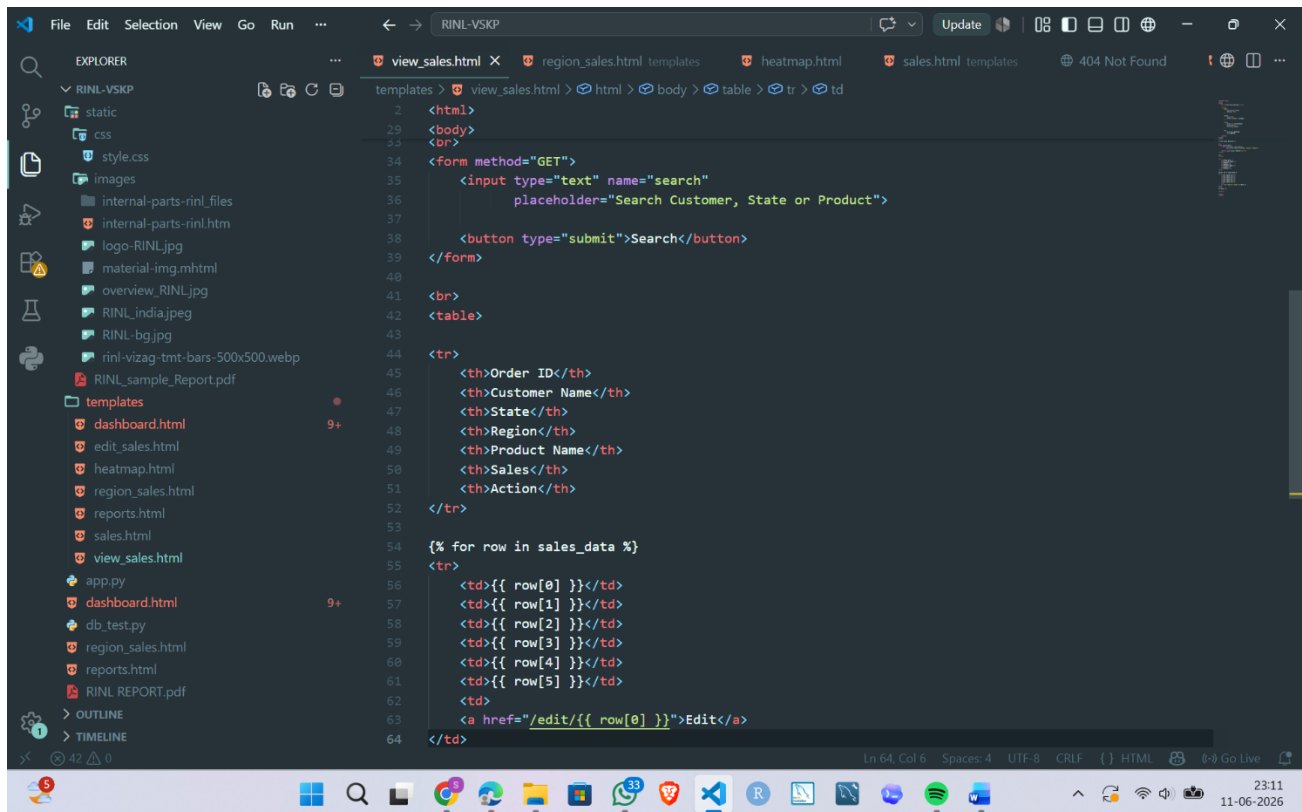
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>RINL Dashboard</title>
5
6 <link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
7 <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
8
9 <style>
10     body{
11         font-family: Arial, sans-serif;
12         background-color: #f4f4f4;
13         margin: 0;
14     }
15
16     .navbar{
17         background-color: #003366;
18         color: white;
19         padding: 15px;
20         text-align: center;
21     }
22
23     .container{
24         padding: 20px;
25     }
26
27     .card{
28         background-color: white;
29         padding: 20px;
30         margin: 10px;
31         border-radius: 10px;
32         box-shadow: 0px 2px 5px gray;
33     }
34
```

SALES ENTRY:



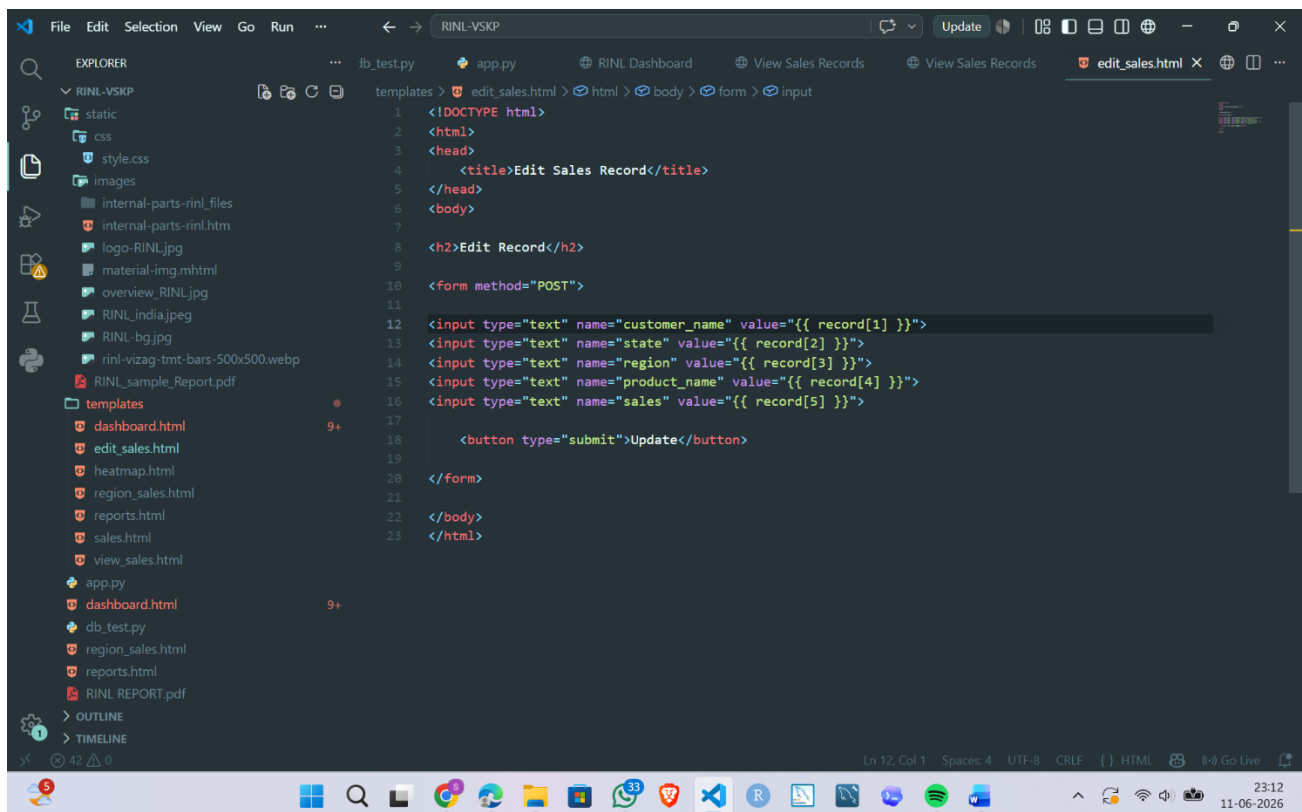
```
1 <html>
2 <body>
3
4 <div class="card">
5
6     <h2>Add Sales Record</h2>
7     {% if message %}
8     <p style="color: green;">{{ message }}</p>
9     {% endif %}
10
11     <form method="POST">
12
13         <input type="text" name="order_id" placeholder="Order ID" required><br><br>
14
15         <input type="date" name="order_date" required><br><br>
16
17         <input type="text" name="customer_name" placeholder="Customer Name" required><br><br>
18
19         <input type="text" name="state" placeholder="State" required><br><br>
20
21         <input type="text" name="region" placeholder="Region" required><br><br>
22
23         <input type="text" name="product_name" placeholder="Product Name" required><br><br>
24
25         <input type="number" step="0.01" name="sales" placeholder="Sales Amount" required><br><br>
26
27         <input type="number" name="quantity" placeholder="Quantity" required><br><br>
28
29         <input type="number" step="0.01" name="profit" placeholder="Profit" required><br><br>
30
31         <button type="submit">Add Sales Record</button>
32
33     </form>
34 </div>
35
```

VIEW SALES:



```
1 <html>
2 <body>
3 <div>
4 <div>
5 <div>
6 <div>
7 <div>
8 <div>
9 <div>
10 <div>
11 <div>
12 <div>
13 <div>
14 <div>
15 <div>
16 <div>
17 <div>
18 <div>
19 <div>
20 <div>
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54 <div>
55 <div>
56 <div>
57 <div>
58 <div>
59 <div>
60 <div>
61 <div>
62 <div>
63 <div>
64 <div>
```

EDIT SALES:



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Edit Sales Record</title>
5 </head>
6 <body>
7 <div>
8 <div>
9 <div>
10 <div>
11 <div>
12 <div>
13 <div>
14 <div>
15 <div>
16 <div>
17 <div>
18 <div>
19 <div>
20 <div>
21 <div>
22 <div>
23 <div>
```

REPORT:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Reports</title>
5 </head>
6 <body>
7
8 <h1>Region Wise Sales Report</h1>
9
10 <table border="1">
11   <tr>
12     <th>Region</th>
13     <th>Total Sales</th>
14   </tr>
15
16   {% for row in report_data %}
17   <tr>
18     <td>{{ row[0] }}</td>
19     <td>{{ row[1] }}</td>
20   </tr>
21   {% endfor %}
22 </table>
23
24 </body>
25 </html>
```

HEATMAP:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>State Wise Sales</title>
5 </head>
6 <body>
7
8 <h1>State Wise Sales</h1>
9
10 <table border="1">
11   <tr>
12     <th>State</th>
13     <th>Sales</th>
14   </tr>
15
16   {% for row in state_sales %}
17   <tr>
18     <td>{{ row[0] }}</td>
19     <td>{{ row[1] }}</td>
20   </tr>
21   {% endfor %}
22 </table>
23
24 </body>
25 </html>
```

CONCLUSION

The **Customer Region-Wise Demand & Sales Analytics System for RINL** was developed to analyze sales data and identify customer demand across different regions and states. The system successfully integrates data management, sales analytics, reporting, and visualization into a single web-based platform.

Using **Python, Flask, MySQL, HTML, CSS, and Chart.js**, the application provides features such as sales entry, sales record management, search and edit functionality, reports, heatmaps, and interactive dashboards. These features help users monitor sales performance, identify high-demand regions, and gain valuable business insights.

Overall, the project demonstrates how data analytics and visualization can transform raw sales data into meaningful information, supporting better decision-making and improving business efficiency. Future enhancements may include data export, user authentication, advanced analytics, and demand forecasting capabilities.

REFERENCES

<https://nodejs.org/en/docs/>

<https://expressjs.com/>

<https://dev.mysql.com/doc/>

<https://www.w3schools.com/>

GLOSSARY

HTML

CSS

PYTHONFLASK

MYSQL

CHART.JS

All the above mentioned are the names of different web scripting languages used in the development.